IQRA UNIVERSITY IU
H-9 CAMPUS, ISLAMABAD
FACULTY OF ENGINEERING, SCIENCES AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY

Course Title	DATABASE MANAGEMENT SYSTEMS LAB		
Course Code:	CSC385L / CSE385L / CSC331L / CMC331L	**Semester:**	Fall 2026
Credit Hours:	1 (Lab), 3 contact hours per week	**Prerequisite(s):**	CMC241-L
Instructor:	Faiza Ali (Junior Lecturer)	**Class:**	BSCS – Section A LAB
Office:	A8	**E-mail:**	faiza.ali@iqraisb.edu.pk
Lab Day:	Tuesday 11:41 – 14:45	**Consultation Hours:**	As announced on LMS
Lab Room:	COMP LAB3	**Knowledge Group:**	Core / Information Management

Course Description:

The course introduces practical database concepts, relational data modeling, database design, SQL, normalization, transaction handling, indexing, backup/recovery, and introductory NoSQL concepts through hands-on laboratory work. Practical work is carried out on Oracle Database (Oracle Database 21c XE / 23ai Free) using Oracle SQL Developer, following the Oracle SQL and PL/SQL course books.

Course Objectives:

The lab develops practical competence in creating and manipulating relational databases, designing schemas and ER models, normalizing tables, executing SQL queries, and applying database management techniques in a DBMS environment.

Course Outline:

Database creation; ERD basics and advanced ERD design; tables and constraints; normalization (1NF, 2NF, 3NF, BCNF); CRUD operations; aggregate functions and grouping; joins; subqueries; stored procedures and triggers; indexing; transactions and rollbacks; open-ended lab/project.

Course academic content is aligned to the Curriculum Book entry for CMC331-L Database Systems Lab.

Course Learning Outcomes (CLOs):

CLO	At the end of the course, the students will be able to:	GA	BT Level*
CLO1	Perform queries as demonstrated and follow instructions to build a database model.	GA 2	P-3
CLO2	Analyze conceptual, logical and physical database schemas using ERD and identify functional dependencies to resolve anomalies for normalized tables.	GA 3	C-4
CLO3	Participate in lab activities and projects individually and as an effective team member to solve real-life database problems.	GA 6	A-3

* BT = Bloom's Taxonomy; C = Cognitive, P = Psychomotor, A = Affective.

Mapping of Course Learning Outcomes (CLOs) to Graduate Attributes (GAs)			
GAs / CLOs	**CLO1**	**CLO2**	**CLO3**
GA1 Academic Education			
GA2 Knowledge for Solving Computing Problems	**X**		
GA3 Problem Analysis		**X**	
GA4 Design/Development of Solutions			
GA5 Modern Tool Usage			
GA6 Individual and Team Work			**X**
GA7 Communication			
GA8 Computing Professionalism and Society			
GA9 Ethics			
GA10 Life-long Learning			

Assessment Modules and Weightages	
Assessment	**Weightage**
Quizzes	0%
Lab Manual / Tasks	15%
Project / OEL / PBL	20%
Mid Semester Examination	25%
End Semester Examination	40%
Total	100%

Books / Reference Material	
1	Fundamentals of Database Management Systems — Mark L. Gillenson, 2nd Edition, 2023, John Wiley & Sons Inc.
2	Modern Database Management — Jeff Hoffer, Ramesh Venkataraman, and Heikki Topi, 13th Edition, 2021.
3	Lab reference: Oracle Database 11g: SQL Fundamentals I (Volumes I and II), Oracle University.
4	Lab reference: Oracle Database 11g: SQL Fundamentals II, Oracle University.
5	Lab reference: Introduction to Oracle9i: PL/SQL (Volumes 1 and 2), Oracle University.

Semester Weekly Plan			
Week No.	**Topic / Contents**	**Lab No.**	**CLO**
Week 1	Introduction to the DBMS environment (Oracle Database, SQL Developer) and database creation	1	CLO1
Week 2	Entity-Relationship Diagram (ERD) basics	2	CLO2
Week 3	Advanced ERD design	3	CLO2

Week 4	Creating tables and constraints	4	CLO1
Week 5	Normalization: 1NF and 2NF	5	CLO2
Week 6	Normalization: 3NF and BCNF	6	CLO2
Week 7	Open Ended Lab / Project / PBL assignment	–	CLO3
Mid Semester Examination (Week 8)			
Week 9	Basic CRUD operations	7	CLO1
Week 10	Aggregate functions and grouping	8	CLO1
Week 11	Joins	9	CLO1
Week 12	Subqueries and nested queries	10	CLO1
Week 13	Stored procedures and triggers	11	CLO1
Week 14	Indexing	12	CLO1
Week 15	Transactions and rollbacks	13	CLO1
Week 16	Open Ended Lab / Project / PBL assessment	–	CLO3
End Semester Examination (Week 17)			

Tools / Hardware / Software Requirement
Oracle Database 21c Express Edition (XE) or Oracle Database 23ai Free; Oracle SQL Developer (SQL client/IDE) or Oracle Live SQL (browser); ER modeling tool (Oracle SQL Developer Data Modeler or draw.io). The SQL taught is standard and can also be practised on SQL Server, PostgreSQL or MySQL with minor syntax changes.

Flexibility of Course Schedule: The order and coverage of topics may be modified based on academic needs, student learning progress, accreditation requirements and emerging technologies while maintaining the intended learning outcomes.

Grading / Academic Conduct Policy	
Quiz Policy:	Quizzes may be conducted either as announced or unannounced assessments. Questions will assess understanding of recently covered concepts. Students are expected to remain up to date with course material.
Assignment / Task Policy:	Assignments and practical tasks must be submitted by the announced deadlines. Late submissions may only be accepted under exceptional circumstances approved by the instructor.
Plagiarism:	All submitted work must reflect the student's own understanding and effort. External sources must be properly cited. Cheating, plagiarism, or unauthorized collaboration may result in zero marks and further disciplinary action under university policy.

Prepared for Fall 2026 | Instructor: Faiza Ali (Junior Lecturer)